Phase portraits for complex Eigenvalues

Now we want to look at the phase portraits of systems with complex Eigenvalues. Here's what we need to know:

1. There are no linear trajectories for the system.

2. The equilibrium of a system with complex Eigenvalues that have no real part is a stable center around which the trajectories revolve, without ever getting closer to or further from equilibrium. The equilibrium of a system with complex Eigenvalues with a positive real part is an unstable spiral that repels all trajectories. The equilibrium of a system with complex Eigenvalues with a negative real part is an asymptotically stable spiral that attracts all trajectories.

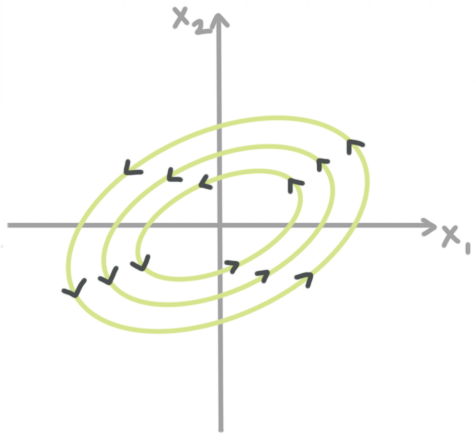
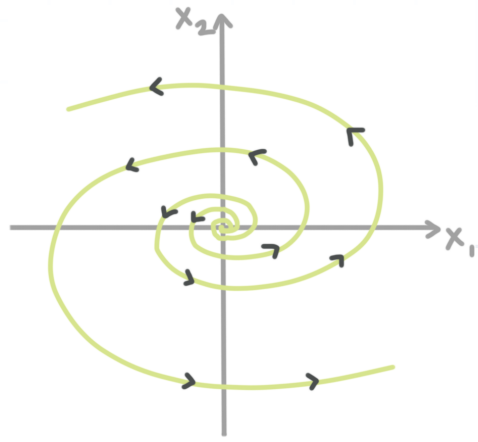
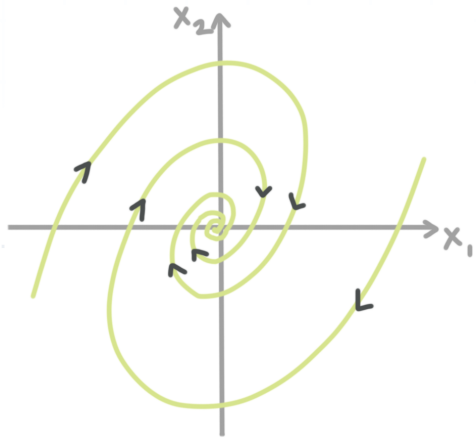
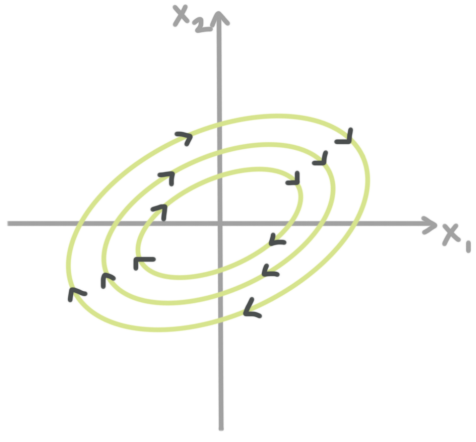
3. To check for the direction of the trajectories for any case of complex conjugate Eigenvalues, we can use the $(1,0)$ test, which means we'll look at the direction at the point $(1,0)$.

Below is a table that summarizes information about the phase portrait, based on the sign of the complex conjugate Eigenvalues.

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COMPLEX CONJUGATE EIGENVALUES

	No real part	Positive real part	Negative real part
	$\lambda_1 = \beta i,\ \lambda_2 = -\beta i$	$\lambda_1, \lambda_2 = \alpha \pm \beta i$	$\lambda_1, \lambda_2 = \alpha \pm \beta i$
		$\alpha > 0$	$\alpha < 0$
Equilibrium	Center	Spiral	Spiral
Stability	Stable	Unstable	Asymptotically stable
		(Repeller)	(Attractor)
Direction	(1,0)	(1,0)	(1,0)
Sketches			
			

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Let's do an example so that we can see how to build a phase portrait for complex conjugate Eigenvalues.

Example

Sketch the phase portrait of the system.

$$x_1' = 4x_1 + 5x_2$$

$$x_2' = -2x_1 + 6x_2$$

This is one of the examples we looked at when we learned to solve homogeneous systems with complex conjugate Eigenvalues, and we found that the Eigenvalues for the system were $\lambda_1 = 5 + 3i$ and $\lambda_2 = 5 - 3i$.

These are Eigenvalues with a positive real part, $\alpha = 5 > 0$, which means we're dealing with an unstable spiral that repels all trajectories.

If we test the vector $\vec{x} = (1,0)$ in the matrix equation associated with the system, we get

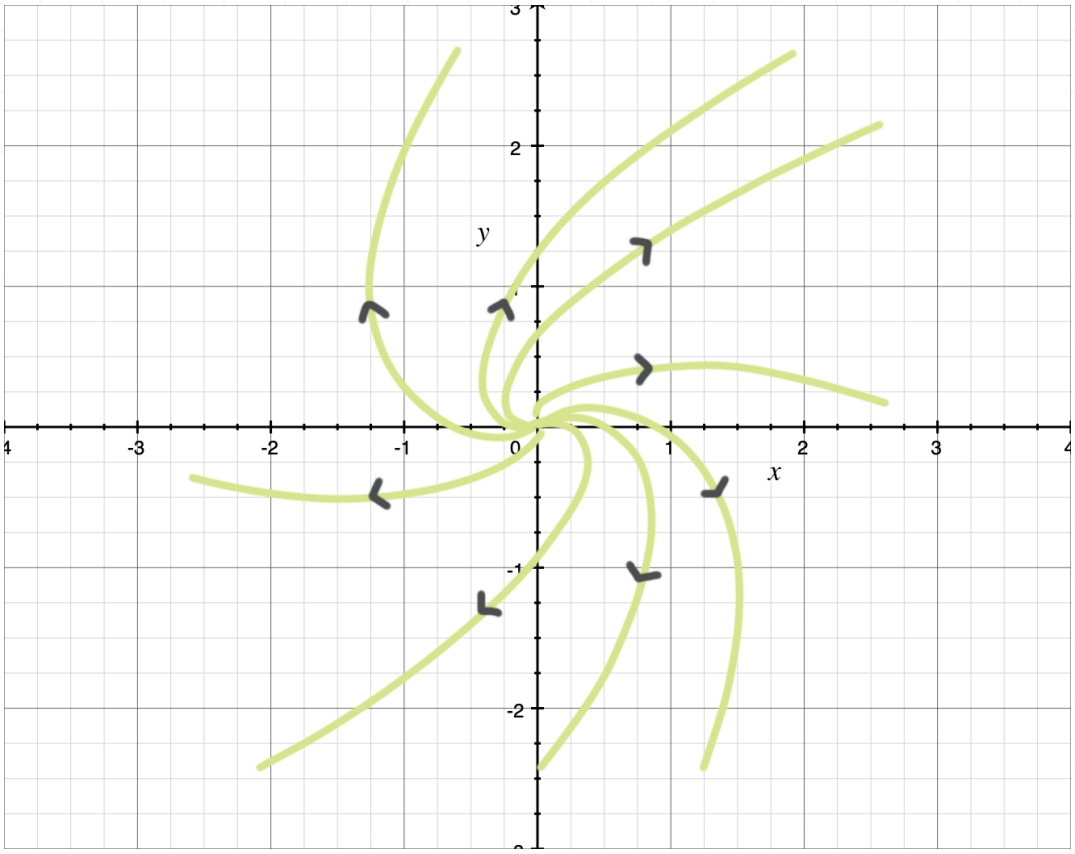
$$\vec{x}' = \begin{bmatrix} 4 & 5 \\ -2 & 6 \end{bmatrix} \vec{x}$$

$$\vec{x}' = \begin{bmatrix} 4 & 5 \\ -2 & 6 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\vec{x}' = \begin{bmatrix} 4 \\ -2 \end{bmatrix}$$

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This $(1,0)$ test tells us that the direction of the trajectory running through that point must have a direction toward $(4, -2)$ (into the fourth quadrant), which means that the phase portrait must look something like



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